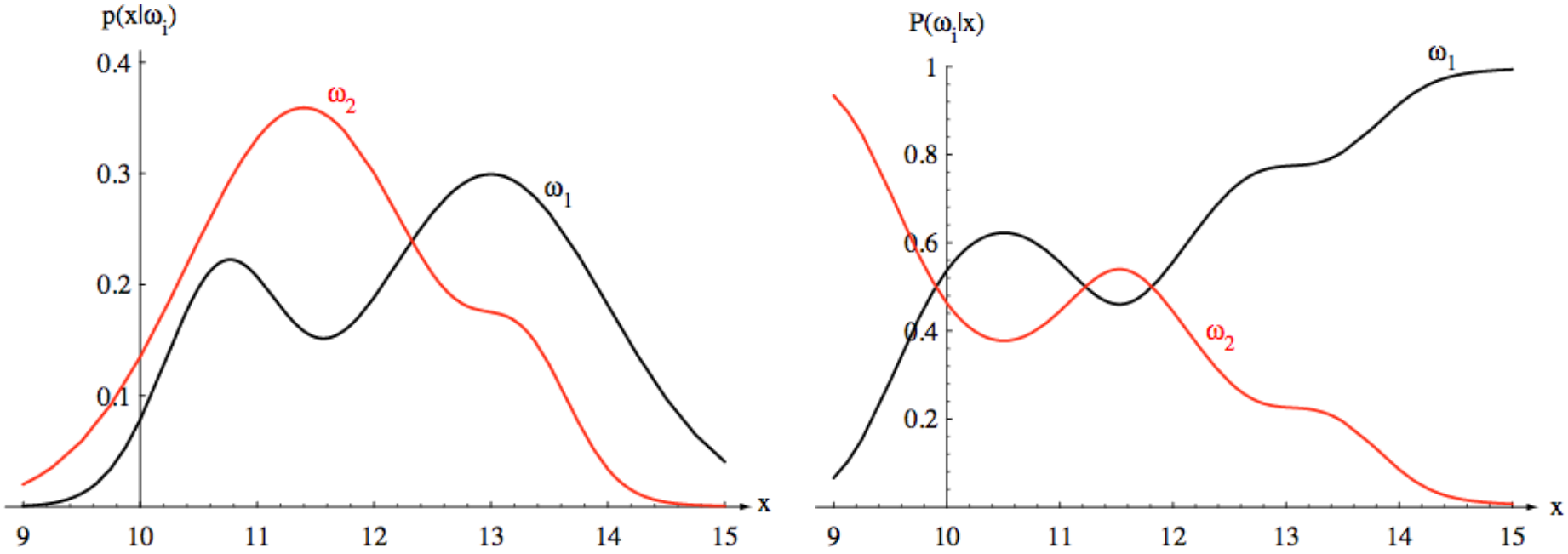
GMM & EM

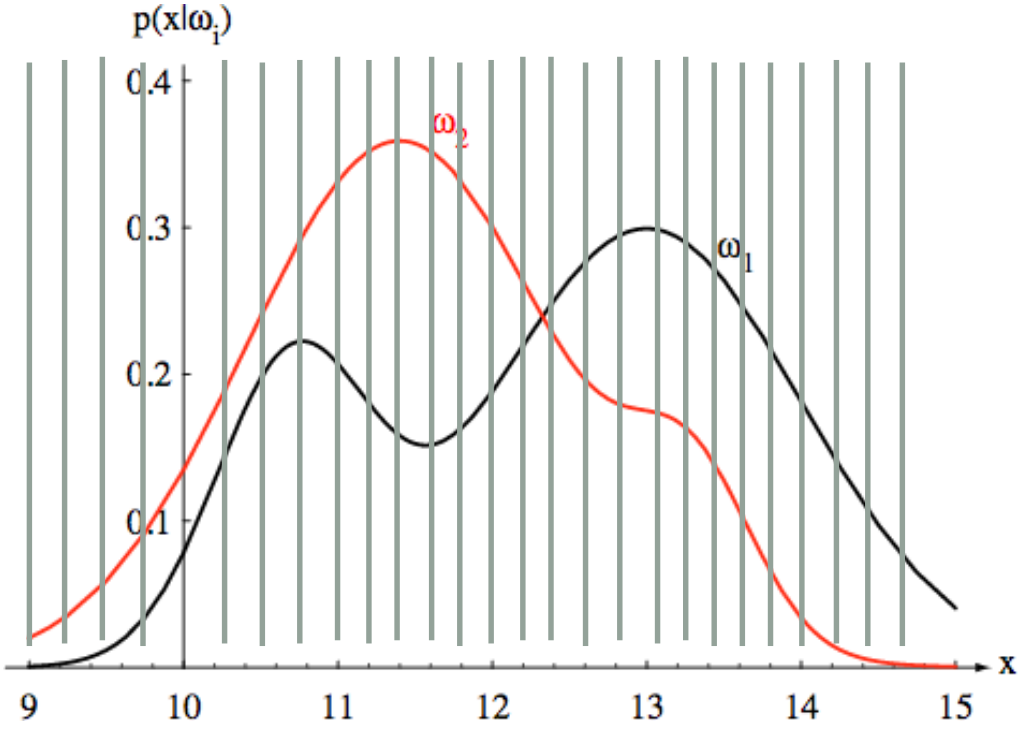
A simple decision rule

- If we can know either p(x|w) or p(w|x) we can make a classification guess



Goal: Find p(x|w) or p(w|x) by finding the parameter of the distribution

A simple way to estimate p(x|w)



Make a histogram!

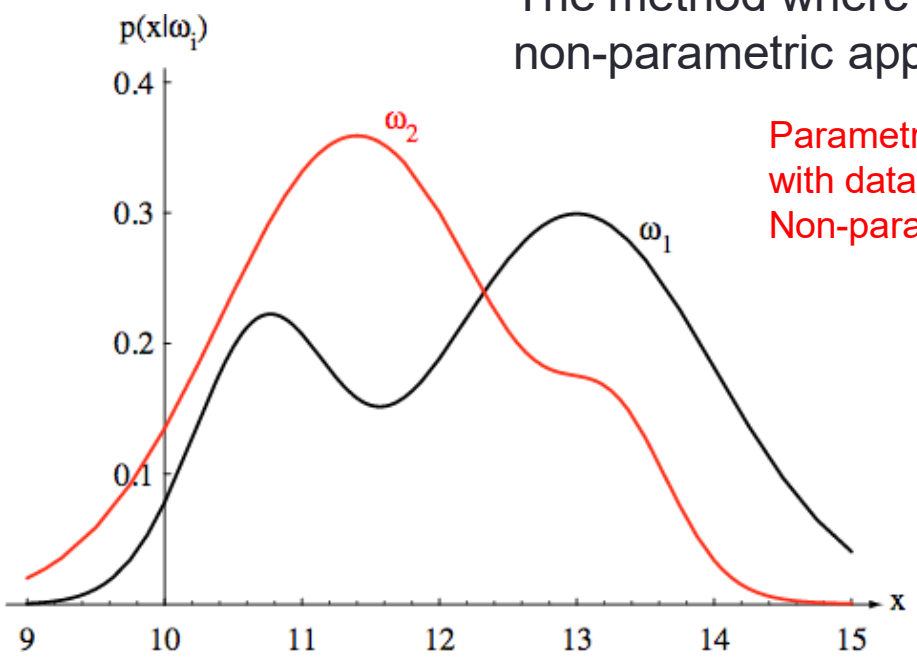
What happens if there is no data in a bin?

The parametric approach

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

- We assume p(x|w) or p(w|x) follow some distributions with parameter θ

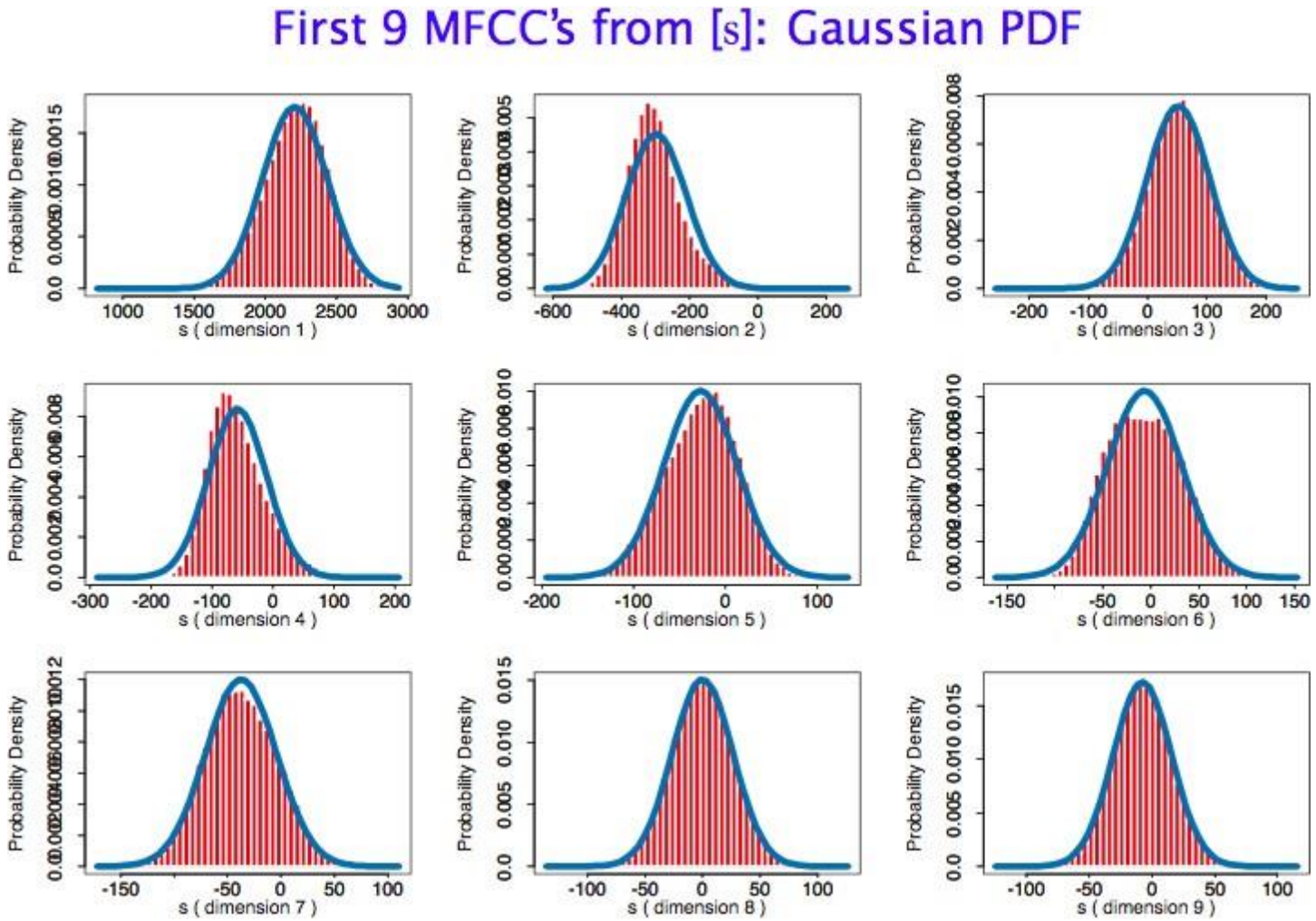
The method where we find the histogram is the non-parametric approach

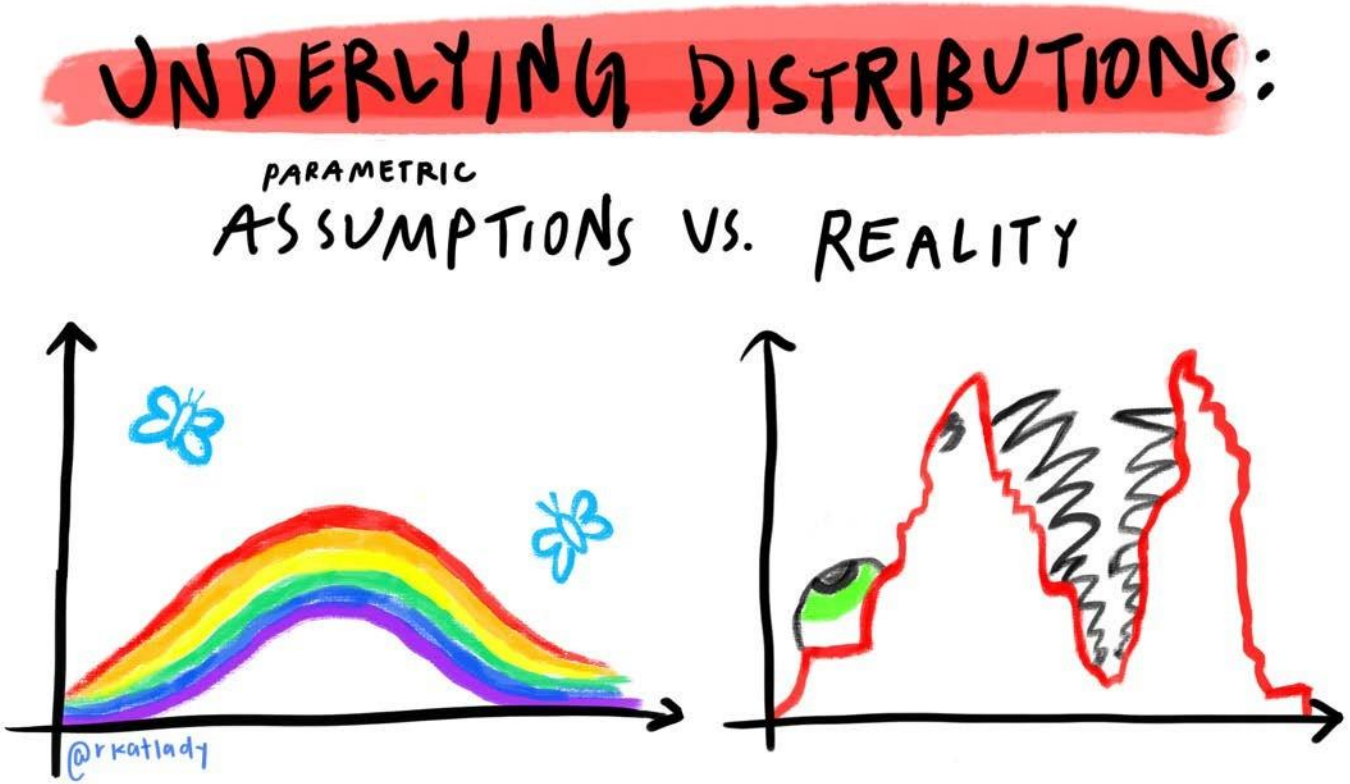


Parametric: need to assume the distribution (might not fit well with data)
Non-parametric: might encounter sparse bins

Goal: Find θ so that we can estimate p(x|w) or p(w|x)

Model of one Gaussian





https://www.khstats.com/blog/tmle/tutorial/

Mixture models

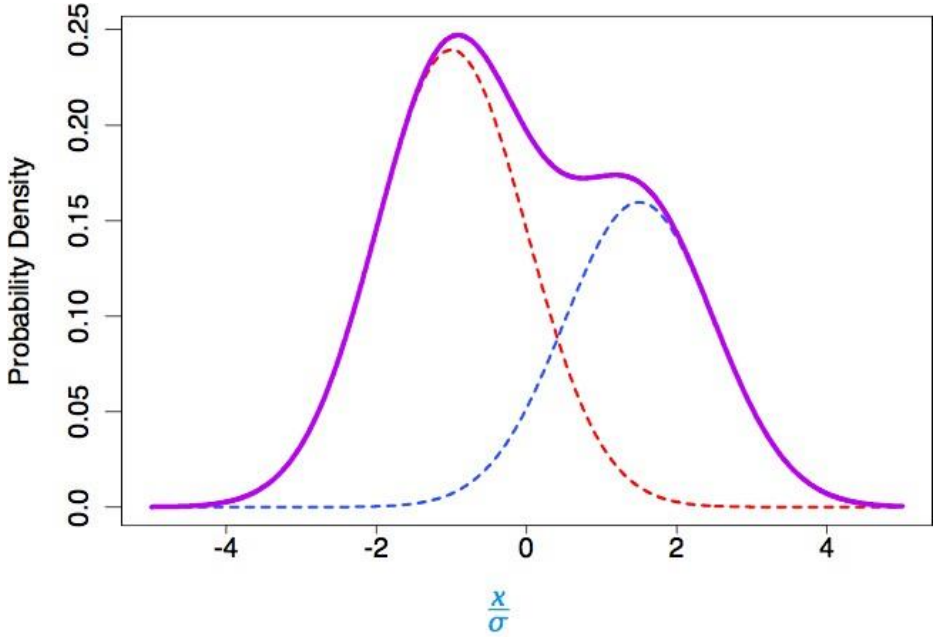
$$p(x) = \sum_k p(k) p_k(x)$$

- A mixture of models from the same distributions (but with different parameters)
- Different mixtures can come from different sub-class
  - Cat class
    - Siamese cats
    - Persian cats
- p(k) is usually categorical (discrete classes)
- Usually the exact class for a sample point is unknown.
  - Latent variable

Gaussian Mixture Models (GMMs)

- Gaussians cannot handle multi-modal data well
- Consider a class can be further divided into additional factors
- Mixing weight makes sure the overall probability sums to 1

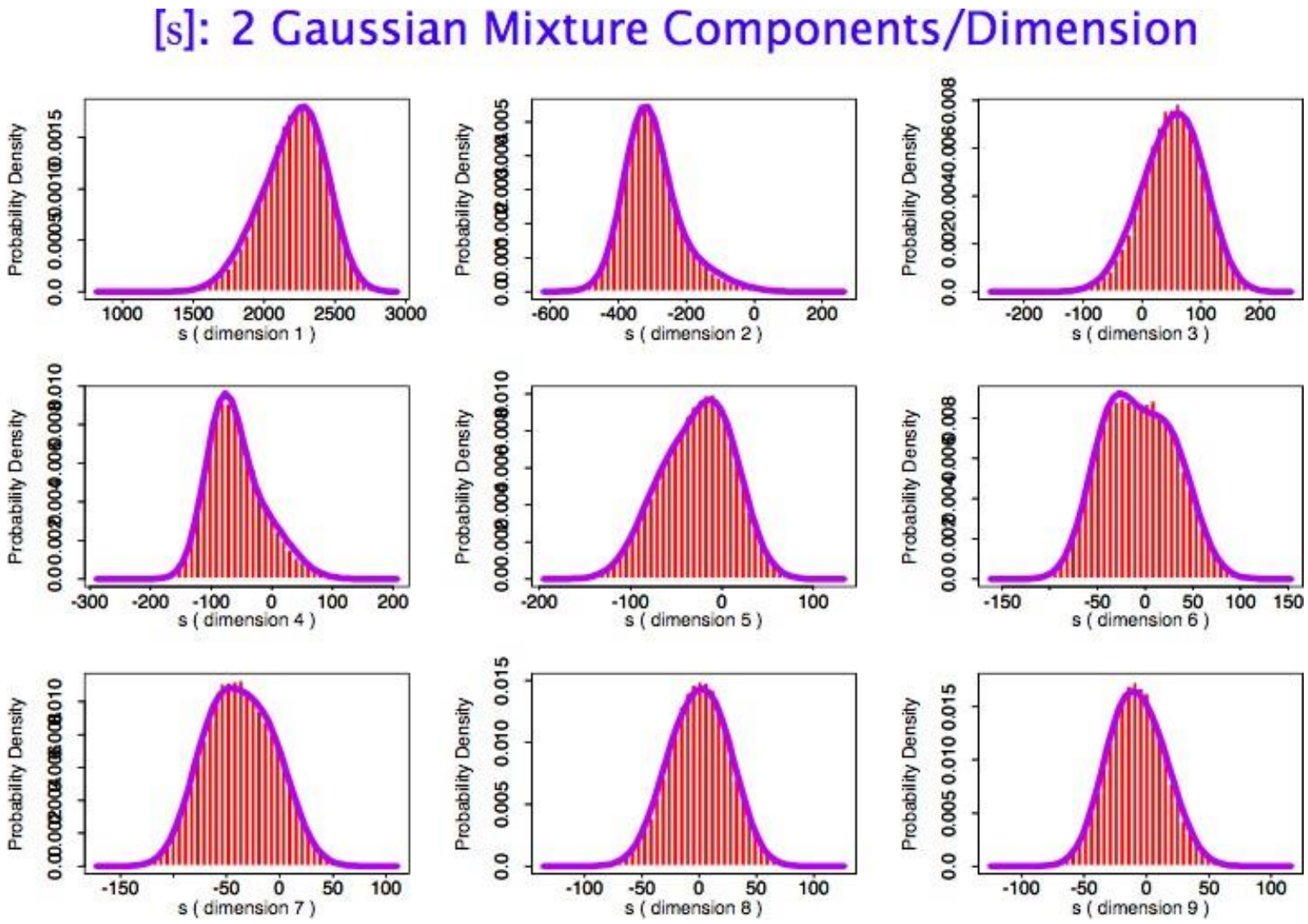
$$P(x) \sim \sum_{k=1}^{K} w_k N(\mu_k, \sigma_k)$$



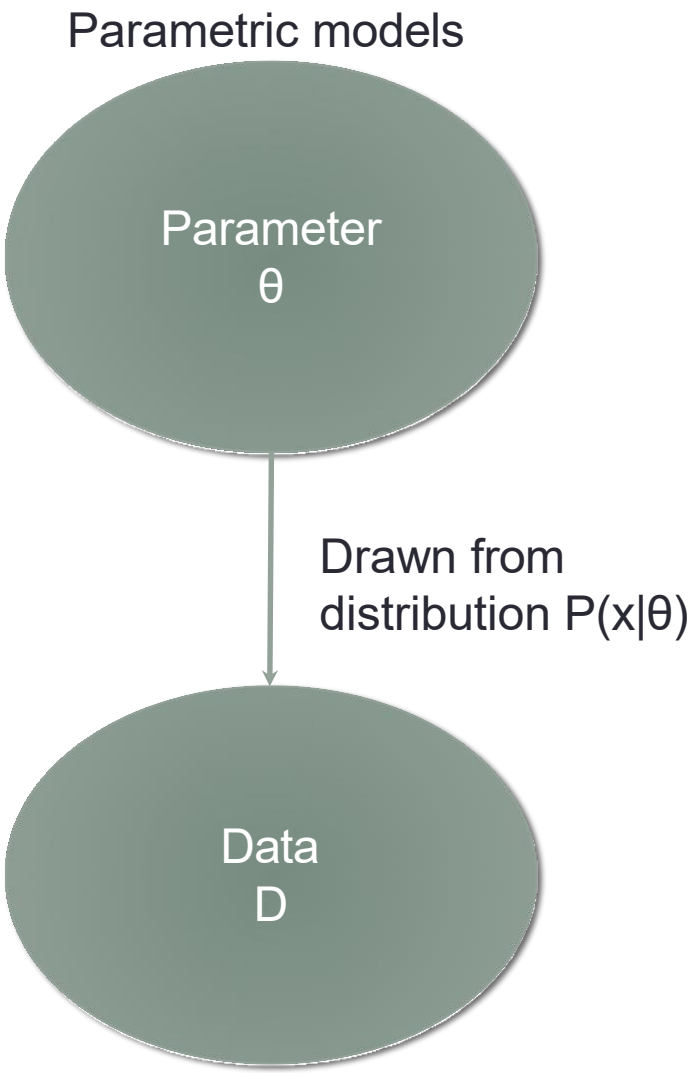
$$p(x) = 0.6p_1(x) + 0.4p_2(x)$$

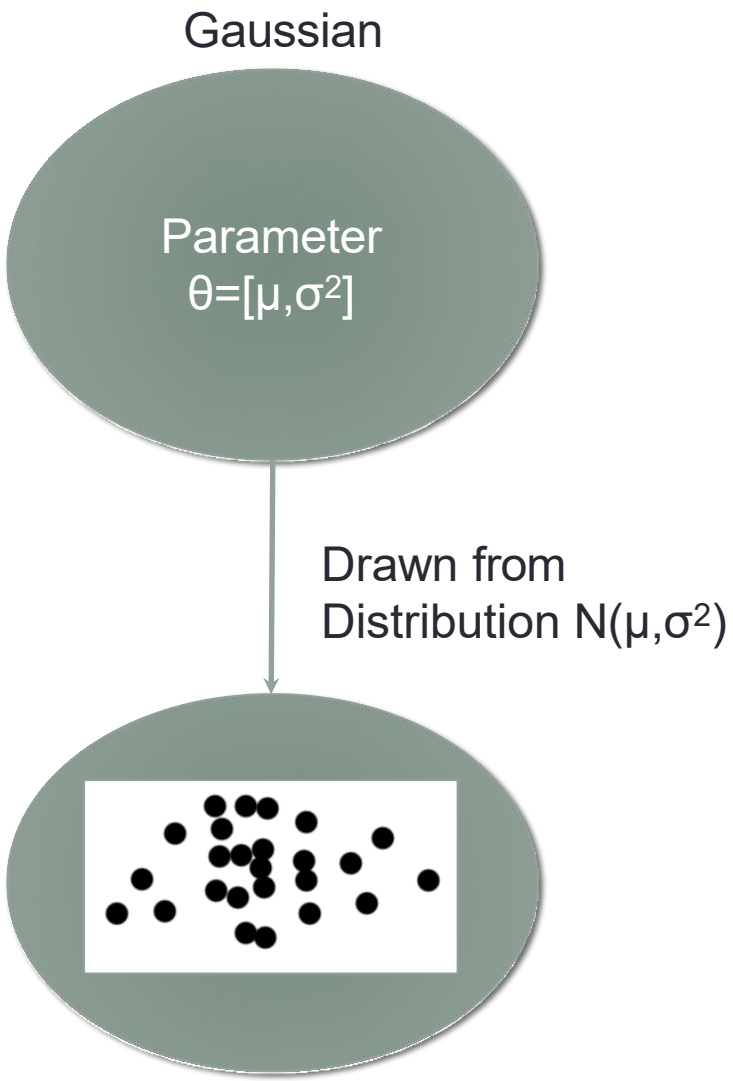
$$p_1(x) \sim N(-\sigma, \sigma^2) \qquad p_2(x) \sim N(1.5\sigma, \sigma^2)$$

Mixture of two Gaussians



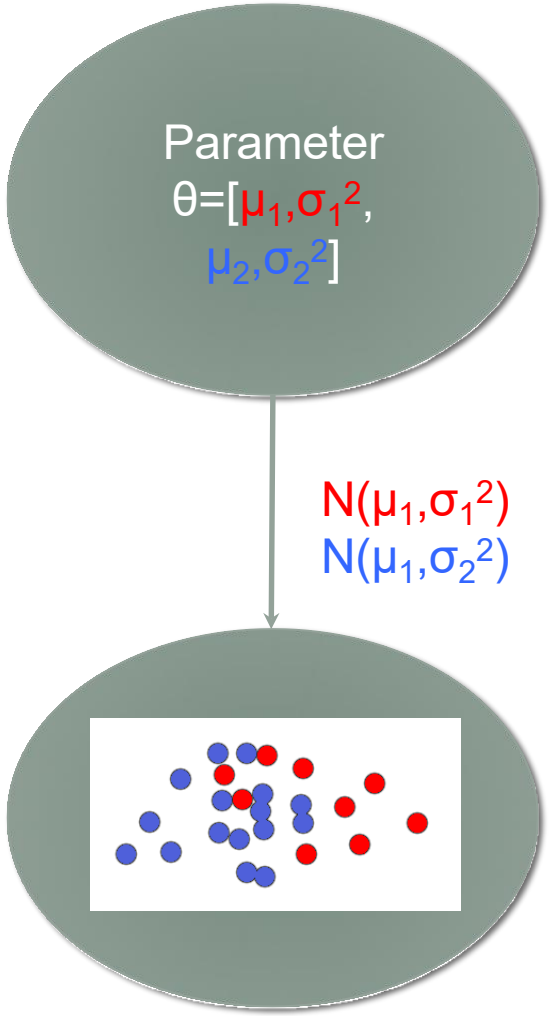
Parametric models



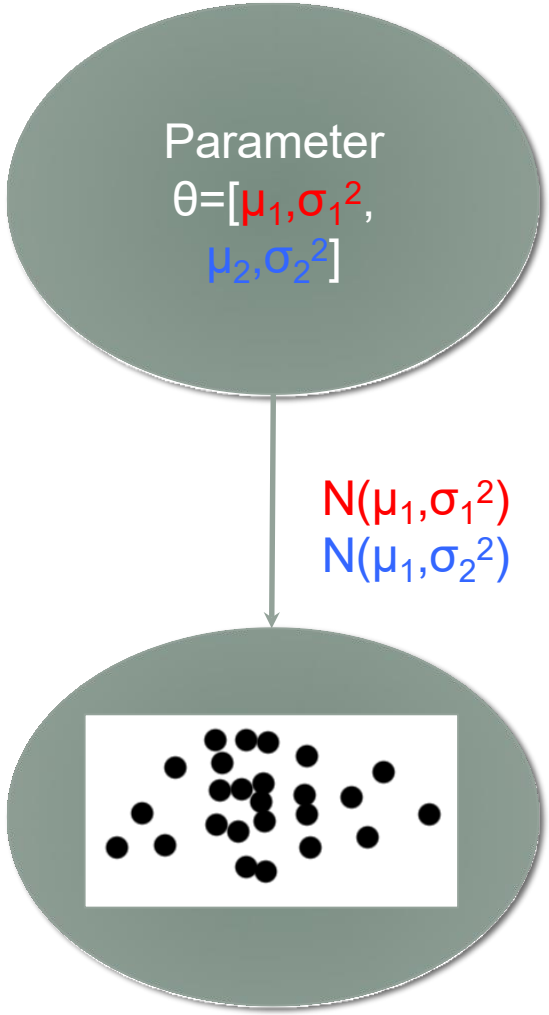


What if some data is missing?

Mixture of Gaussian

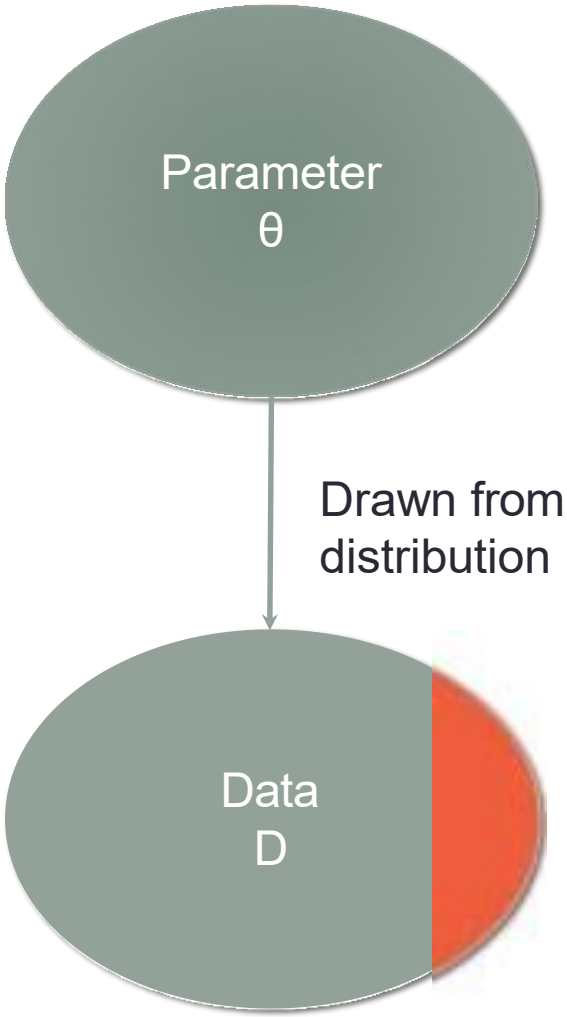


Unknown mixture labels



Estimating missing data

Parametric models



Latent variables,k
unknown

Need to estimate both the latent
Variables and the model parameters.

Slight difference in notation

$p(\mathbf{x}|\theta)$ vs $p(\mathbf{x};\theta)$

θ as a RV at a fixed value vs θ as a fixed parameter

Most of the time can be used interchangeably

Estimating latent variables and model parameters

- GMM $$p(x) = \sum_k p(k) N(\mu_k, \sigma_k)$$
- Observed $(x_1, x_2, \ldots, x_N)$
- Latent $(k_1, k_2, \ldots, k_N)$ from K possible mixtures
- Parameter for p(k) is ϕ , $p(k = 1) = \phi_1$, $p(k = 2) = \phi_2 \ldots$

$$l(\phi, \mu, \Sigma) = \Sigma_{n=1}^{N} log p(x^{(\overset{n}{\cancel{i}})}; \phi, \mu, \sigma)$$

$$= \Sigma_{n=1}^{N} log \boxed{\Sigma_{l=1}^{K}} p(x_n | k_{n,l}; \mu, \sigma) p(k_{n,l}; \phi)$$

Make things hard to solve

Cannot be solved by differentiating

Assuming k

- What if we somehow know k_n?
- Maximizing wrt to ϕ, μ, σ gives

$$\phi_j = \frac{1}{N}\Sigma_{n=1}^{N} 1(k_n = j)$$

$$\mu_j = \frac{\Sigma_{n=1}^{N} 1(k_n = j)x_n}{\Sigma_{n=1}^{N} 1(k_n = j)}$$

$$\sigma_j^2 = \frac{\Sigma_{n=1}^{N} 1(k_n = j)(x_n - \mu_j)^2}{\Sigma_{n=1}^{N} 1(k_n = j)}$$

$$1(condition)$$

Indicator function. Equals one if condition is met. Zero otherwise

Iterative algorithm

- Initialize ϕ, μ, σ
- Repeat till convergence
  - Expectation step (E-step) : Estimate the latent labels **k**
  - Maximization step (M-step) : Estimate the parameters ϕ, μ, σ given the latent labels
- Called Expectation Maximization (EM) Algorithm
- How to estimate the latent labels?

Iterative algorithm

- Initialize ϕ, μ, σ
- Repeat till convergence
  - Expectation step (E-step) : Estimate the latent labels **k** by finding the pdf of k given everything else $p(k| x ; \phi, \mu, \sigma)$
  - Maximization step (M-step) : Estimate the parameters ϕ, μ, σ given the latent labels by maximizing the expectation of the log likelihood
- Extension of MLE for latent variables
  - MLE : argmax $\log p(x;\theta)$
  - EM : argmax $\log \Sigma_k p(x, k;\theta)$

How to evaluate $\log \Sigma_k p(x, k;\theta)$ when we don't know k?

Convex functions and Jensen's inequality

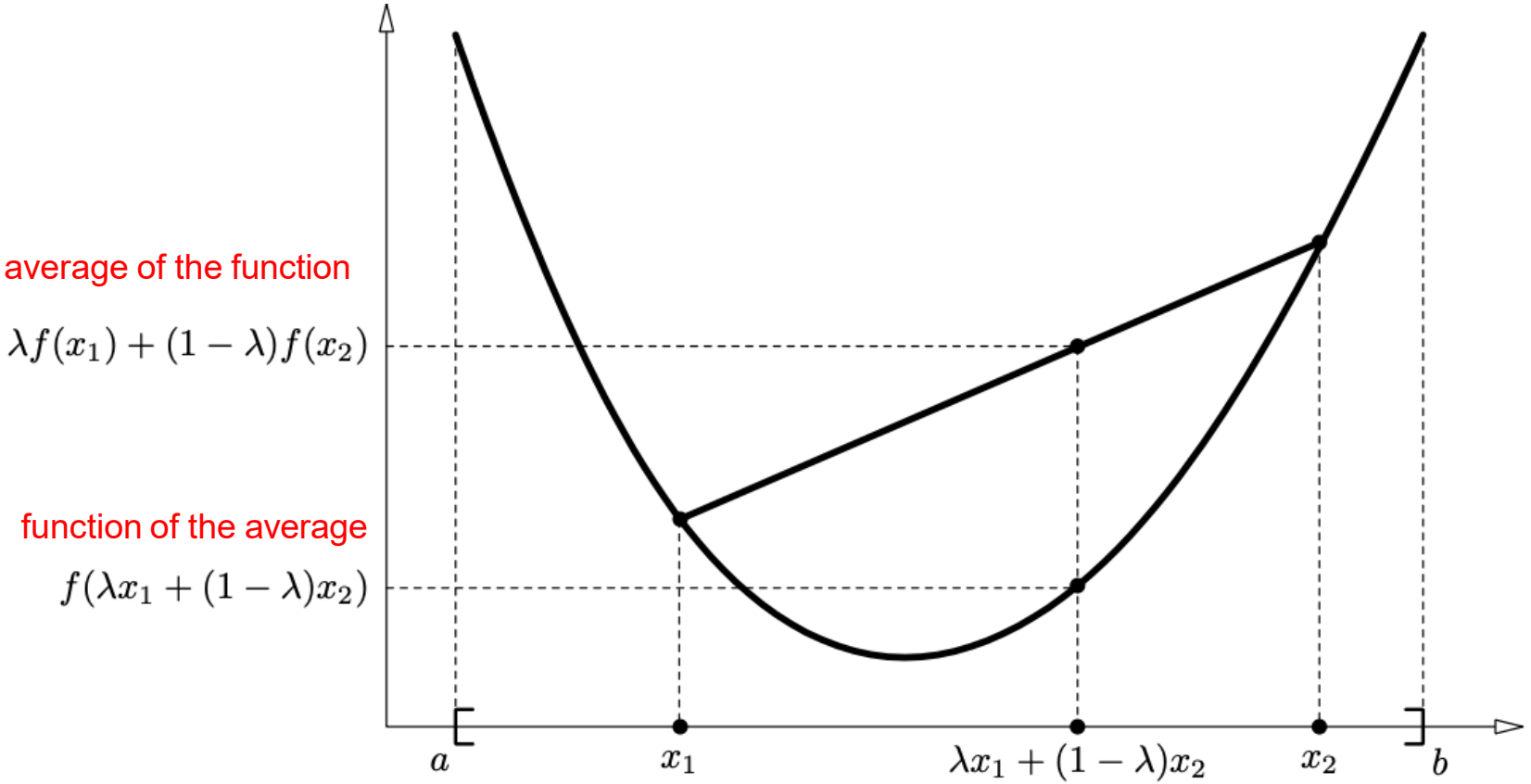


Figure 1: f is *convex* on $[a, b]$ if $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$ $\forall x_1, x_2 \in [a, b], \quad \lambda \in [0, 1]$.

Difference of convexity and strict convexity → https://math.stackexchange.com/questions/2780443/difference-of-convexity-and-strict-convexity

Jensen's inequality

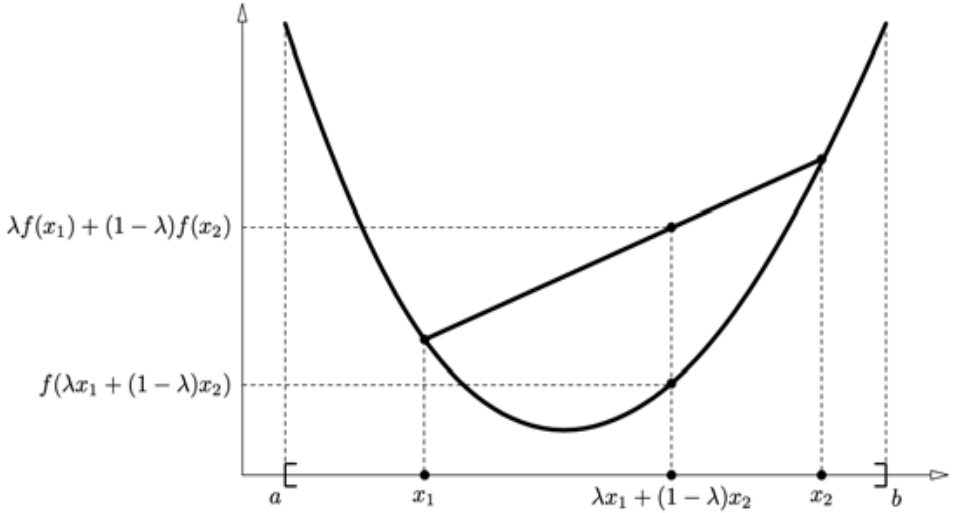


Figure 1: f is *convex* on $[a, b]$ if $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$ $\forall x_1, x_2 \in [a, b], \quad \lambda \in [0, 1]$.

Let f be a convex function on interval I

If $x_1, x_2, \ldots, x_n$ is in I,

$w_1, \ldots, w_n > 0$ and sums to 1

then,

$$f\left(\sum_i^n w_i x_i\right) \leq \sum_i^n w_i f(x_i)$$

If f is concave, flip the inequality.

Can view this as expectation

$$f(E[X]) \leq E[f(X)]$$

Jensen's inequality and ELBO

log Σ_k p(x, k|θ)

$$f(\sum_i^n w_i x_i) \leq \sum_i^n w_i f(x_i)$$

Maximize Evidence Lower Bound (ELBO) = Σ_k Q(k) log (p(x, k;θ)/Q(k))

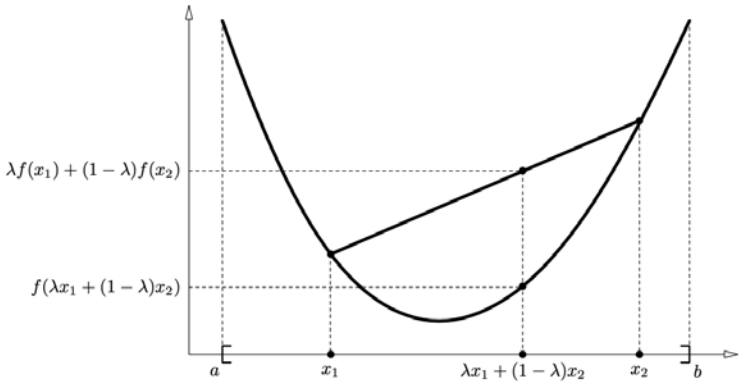
Making the lower bound tight

We will make the bound tight for fixed θ

Jensen's inequality is tight when?

$$f(\sum_i^n w_i x_i) \leq \sum_i^n w_i f(x_i)$$

$$f(E[X]) \leq E[f(X)]$$



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Making the lower bound tight

We will make the bound tight for fixed θ

$$f(\sum_i^n w_i x_i) \leq \sum_i^n w_i f(x_i)$$

$$f(E[X]) \leq E[f(X)]$$

If f( ) is strictly convex, Jensen's inequality is tight IFF

x_i are all equal

E[X] = X = constant

Making the lower bound tight

We will make the bound tight for fixed θ

Jensen's inequality is tight when the inside of the expectation is a constant, c wrt the expectation

$p(x, k;\theta)/Q(k) = c$

or $Q(k) = p(k \mid x; \theta)$

Iterative algorithm (general)

- Goal of EM : $\log \sum_k p(x, k;\theta) >= \Sigma_k Q(k) \log (p(x, k;\theta)/Q(k))$
- Maximize the ELBO instead
- Initialize Θ
- Repeat till convergence
  - Expectation step (E-step) : estimate the conditional expectation $Q(k) = p(k|x;\theta)$ using the current θ.
  - Maximization step (M-step) : Estimate new Θ given by maximizing the ELBO given current $Q(k)$

Will this work?

For iteration i, with $\theta^{(i)}$

ELBO

$\log \sum_k p(x, k;\theta^{(i)}) >= \boxed{\Sigma_k Q(k) \log (p(x, k;\theta^{(i)})/Q(k))}$

E-step, making the bound tight by picking Q'(k) yields

$\log \sum_k p(x, k;\theta^{(i)}) = \Sigma_k Q'(k) \log (p(x, k;\theta^{(i)})/Q'(k))$

M-step, maximize ELBO by finding $\theta^{(i+1)}$

$\Sigma_k Q'(k) \log (p(x, k;\theta^{(i)})/Q'(k)) <= \Sigma_k Q'(k) \log (p(x, k;\theta^{(i+1)})/Q'(k))$

For iteration i+1, with $\theta^{(i+1)}$

$\log \sum_k p(x, k;\theta^{(i+1)}) >= \Sigma_k Q(k) \log (p(x, k;\theta^{(i+1)})/Q(k))$

Thus,

$\log \sum_k p(x, k;\theta^{(i+1)}) >= \log \sum_k p(x, k;\theta^{(i)})$

So EM improves the likelihood at every step!

Notes on ELBO

We set $Q(k) = p(k \mid x; \theta)$ to make the inequality tight.

What if we cannot compute $p(k \mid x; \theta)$?

Use a looser bound by picking any $Q(k)$

Estimate $p(k \mid x; \theta)$ with $q(k \mid x; \theta)$ that we can compute

This is called Variational Inference

We will revisit this.

Estimating latent variables and model parameters

- GMM $$p(x) = \sum_k p(k) N(\mu_k, \sigma_k)$$
- Observed $(x_1, x_2, \ldots, x_N)$
- Latent $(k_1, k_2, \ldots, k_N)$ from K possible mixtures
- Parameter for p(k) is ϕ , $p(k = 1) = \phi_1$, $p(k = 2) = \phi_2 \ldots$

$$l(\phi, \mu, \Sigma) = \Sigma_{n=1}^{N} log p(x^{(i)}; \phi, \mu, \sigma)$$

$$= \Sigma_{n=1}^{N} log \boxed{\Sigma_{l=1}^{K}} p(x_n | k_{n,l}; \mu, \sigma) p(k_{n,l}; \phi)$$

Make things hard to solve

Cannot be solved by differentiating

EM on GMM

- E-step
  - Set soft labels: $w_{n,j}$ = probability that nth sample comes from jth mixture p
  - Using Bayes rule
    - $p(k|x\,;\,\mu,\sigma,\phi) = \dfrac{p(x|k\,;\,\mu,\sigma,\phi)\,p(k;\,\mu,\sigma,\phi)}{p(x;\,\mu,\sigma,\phi)}$
    - $p(k|x\,;\,\mu,\sigma,\phi)$ is proportional to $p(x|k\,;\,\mu,\sigma,\phi)\,p(k;\,\phi)$

$$N(\mu_j, \sigma_j) \qquad \phi_j$$

$$p(k_n = j | x_n; \phi, \mu, \Sigma) = \frac{p(x_n; \mu_j, \sigma_j) p(k_n = j; \phi)}{\Sigma_l p(x_n; \mu_l, \sigma_l) p(k_n = l; \phi)}$$

EM on GMM

- M-step (hard labels)
- we somehow know k_n?

$$\phi_j = \frac{1}{N}\Sigma_{n=1}^{N} 1(k_n = j)$$

$$\mu_j = \frac{\Sigma_{n=1}^{N} 1(k_n = j)x_n}{\Sigma_{n=1}^{N} 1(k_n = j)}$$

$$\sigma_j^2 = \frac{\Sigma_{n=1}^{N} 1(k_n = j)(x_n - \mu_j)^2}{\Sigma_{n=1}^{N} 1(k_n = j)}$$

- M-step (soft labels)

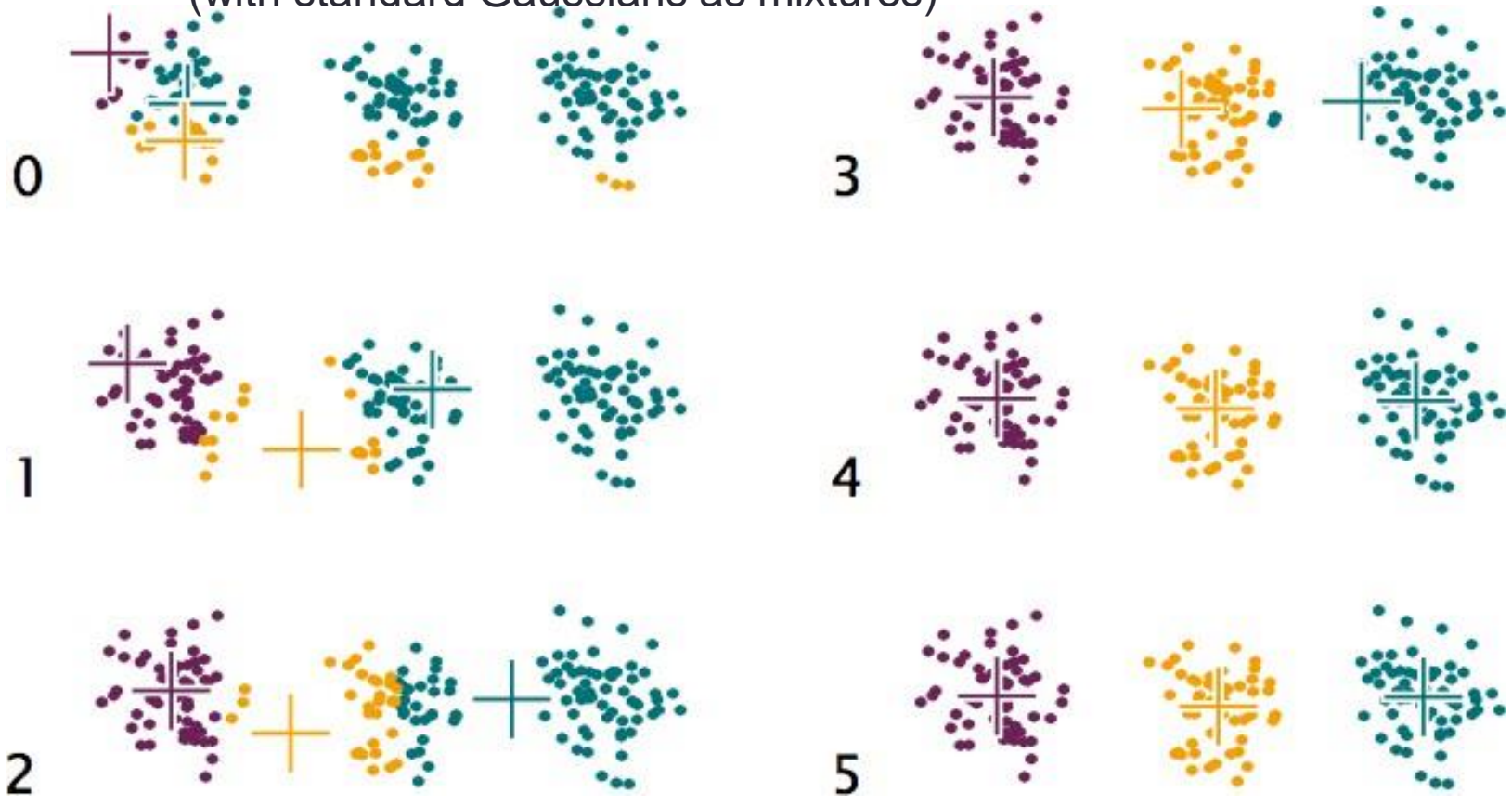
$$\phi_j = \frac{1}{N}\Sigma_{n=1}^{N} w_{n,j}$$

$$\mu_j = \frac{\Sigma_{n=1}^{N} w_{n,j} x_n}{\Sigma_{n=1}^{N} w_{n,j}}$$

$$\sigma_j^2 = \frac{\Sigma_{n=1}^{N} w_{n,j}(x_n - \mu_j)^2}{\Sigma_{n=1}^{N} w_{n,j}}$$

K-mean vs EM

EM on GMM can be considered as EM with soft labels
(with standard Gaussians as mixtures)



K-mean clustering

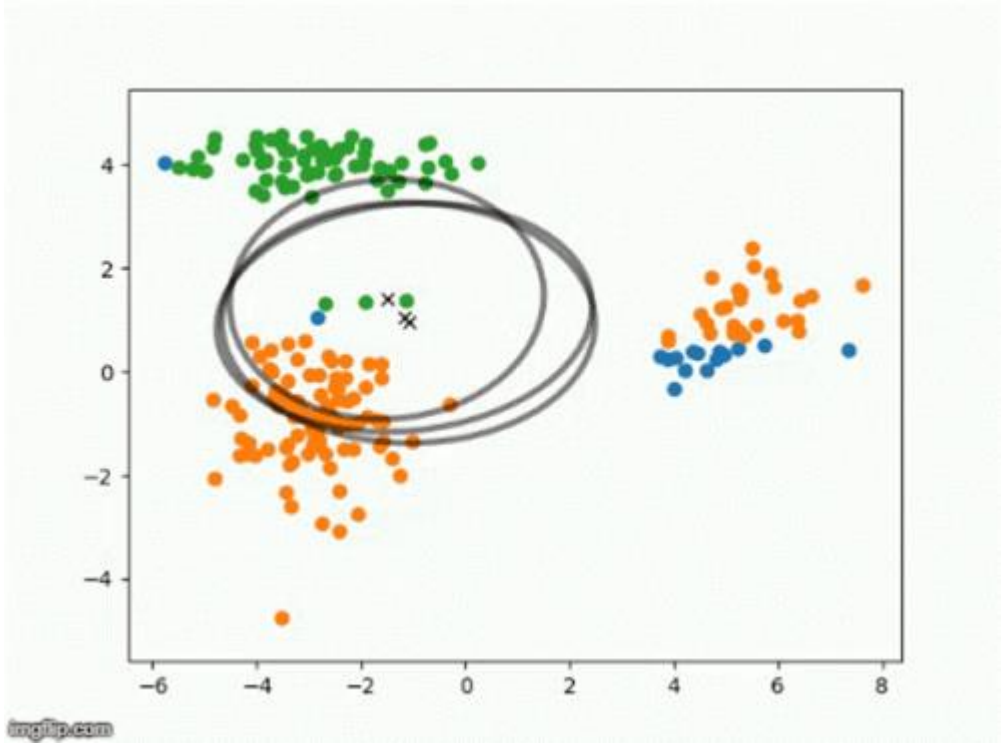
- Task: cluster data into groups
- K-mean algorithm
  - Initialization: Pick K data points as cluster centers
  - Assign: Assign data points to the closest centers
  - Update: Re-compute cluster center
  - Repeat: Assign and Update

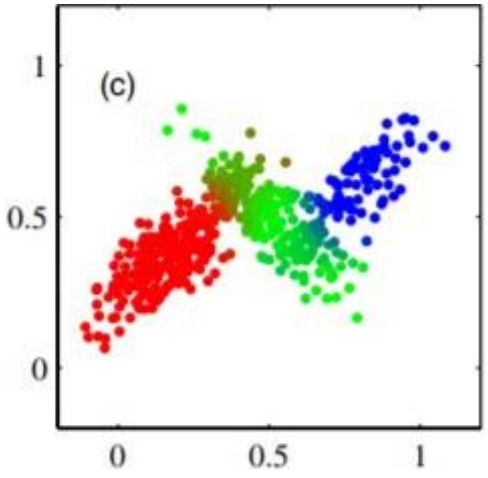
EM algorithm for GMM

- Task: cluster data into Gaussians
- EM algorithm
  - Initialization: Randomly initialize parameters Gaussians
  - Expectation: Assign data points to the closest Gaussians
  - Maximization: Re-compute Gaussians parameters according to assigned data points
  - Repeat: Expectation and Maximization

- Note: assigning data points is actually a soft assignment (with probability)

EM Clustering



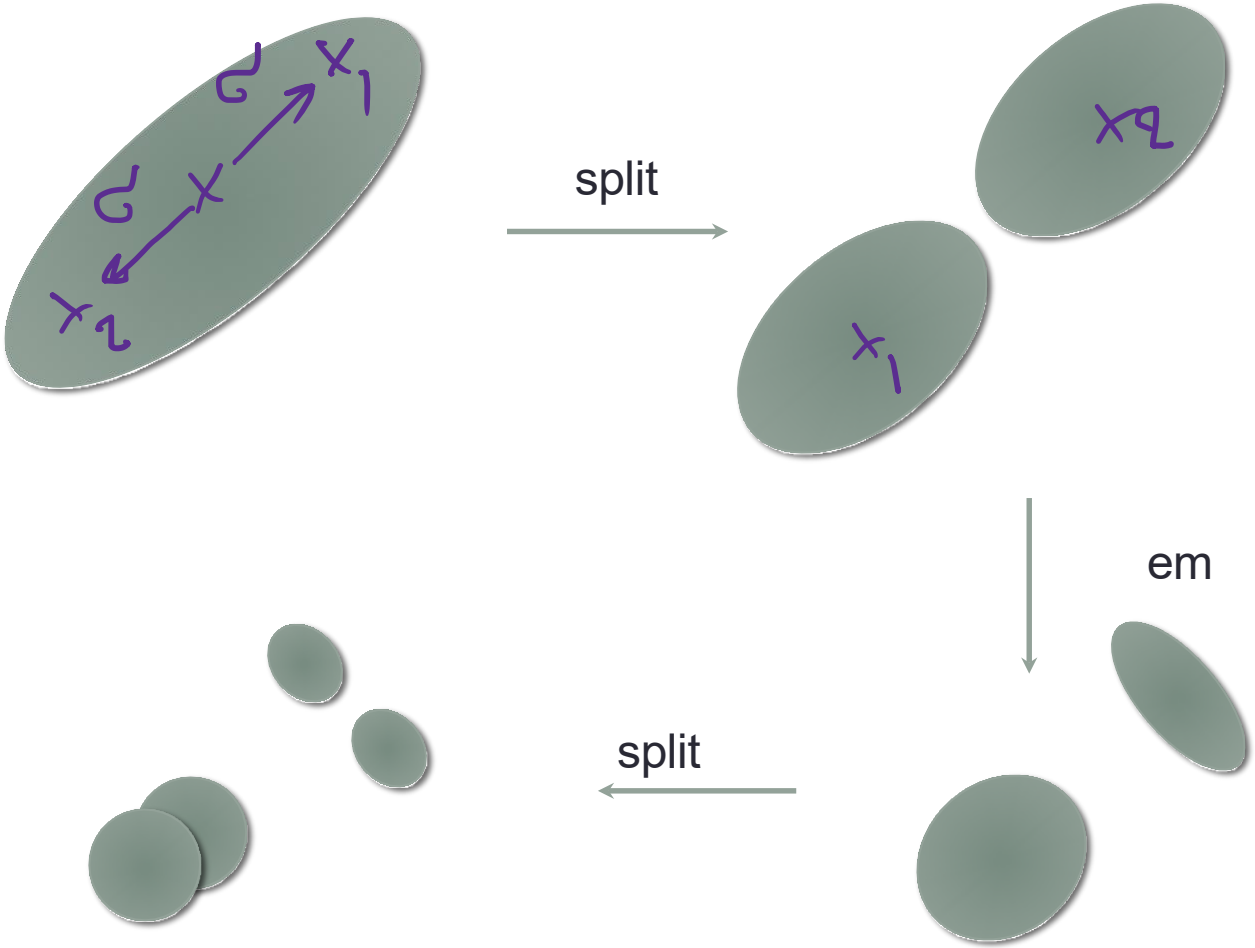


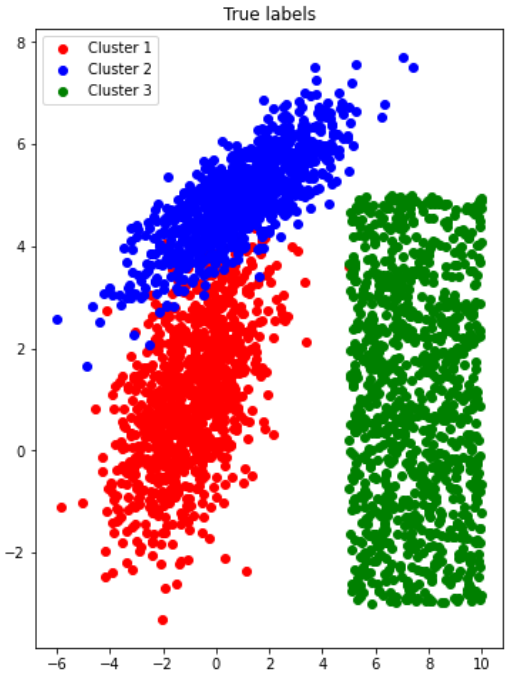
https://towardsdatascience.com/gaussian-mixture-models-vs-k-means-which-one-to-choose-62f2736025f0

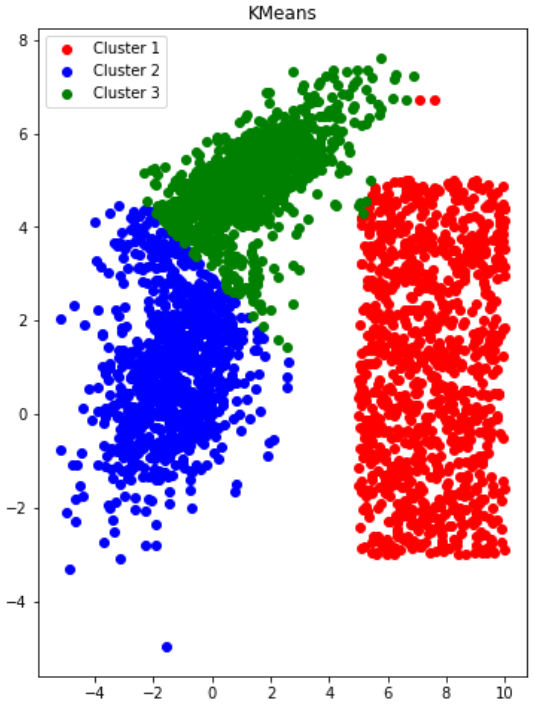
EM/GMM notes

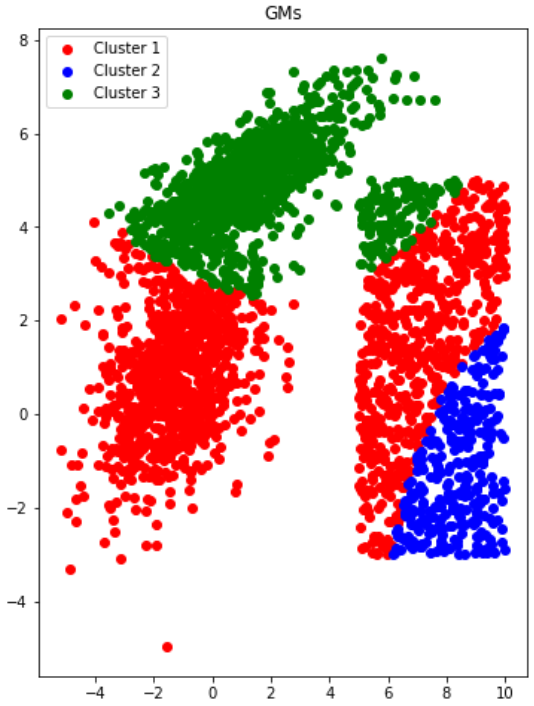
- Converges to local maxima (maximizing likelihood)
  - Just like k-means, need to try different initialization points → Compare likelihood with same k
- EM always improve the likelihood for each iteration
  - Stops EM when likelihood changes < threshold
- Just like k-means some centroid can get stuck with one sample point and no longer moves
  - For EM on GMM this cause variance to go to 0…
    - Introduce variance floor (minimum variance a Gaussian can have)
- Tricks to avoid bad local maxima
  - Starts with 1 Gaussian
  - Split the Gaussians according to the direction of maximum variance
  - Repeat until arrive at k Gaussians
  - Does not guarantee global maxima but works well in practice

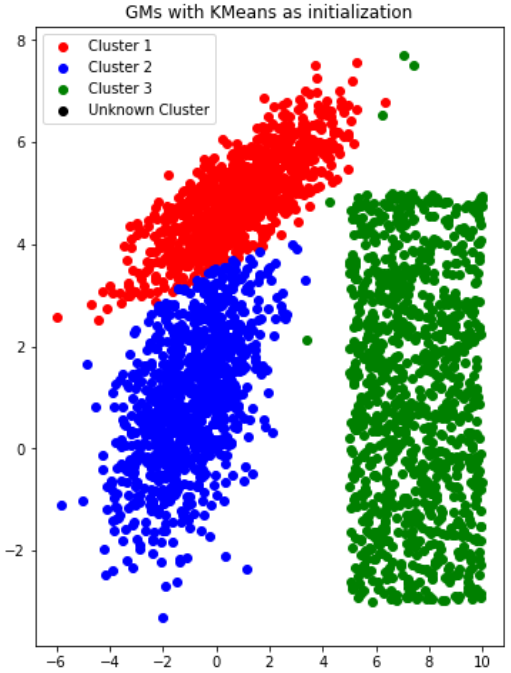
Gaussian splitting by variance







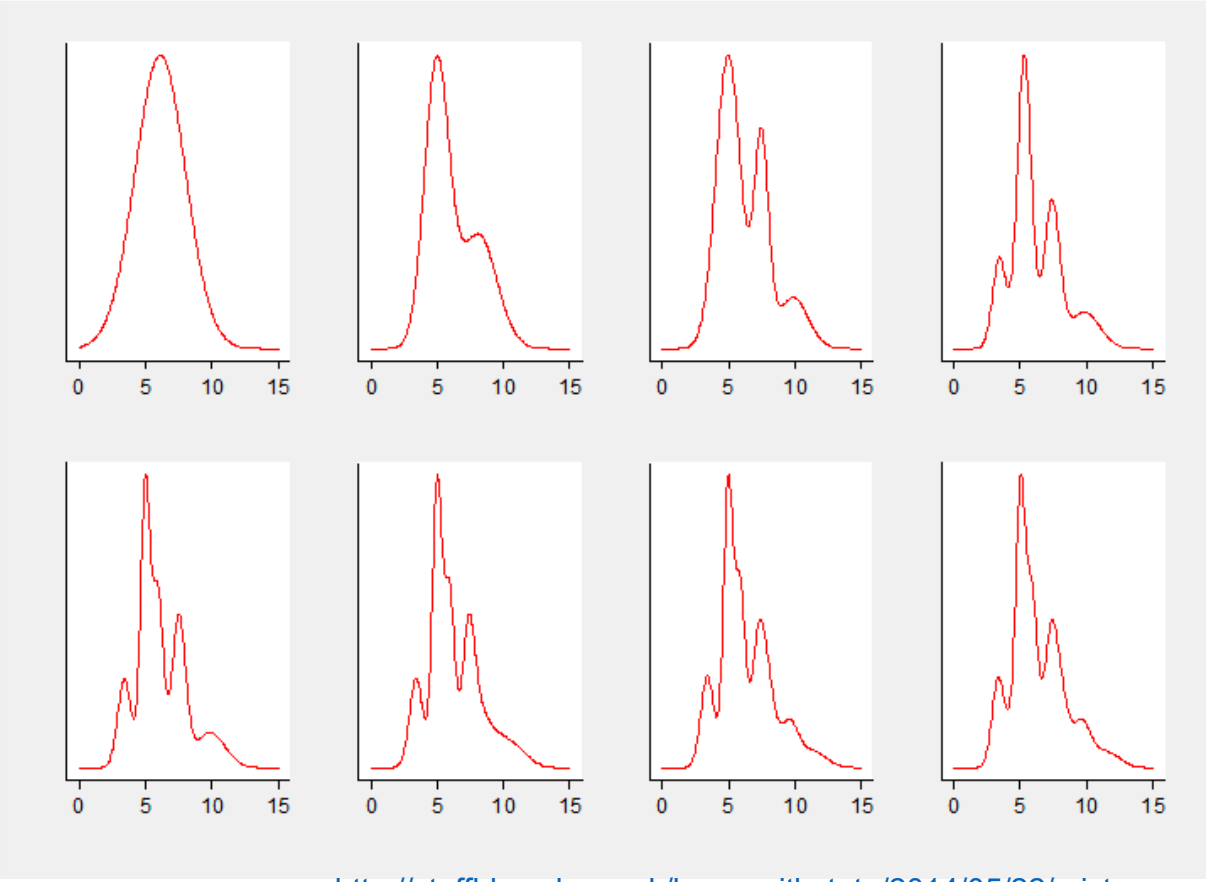




https://towardsdatascience.com/gaussian-mixture-models-vs-k-means-which-one-to-choose-62f2736025f0

Picking the amount of Gaussians

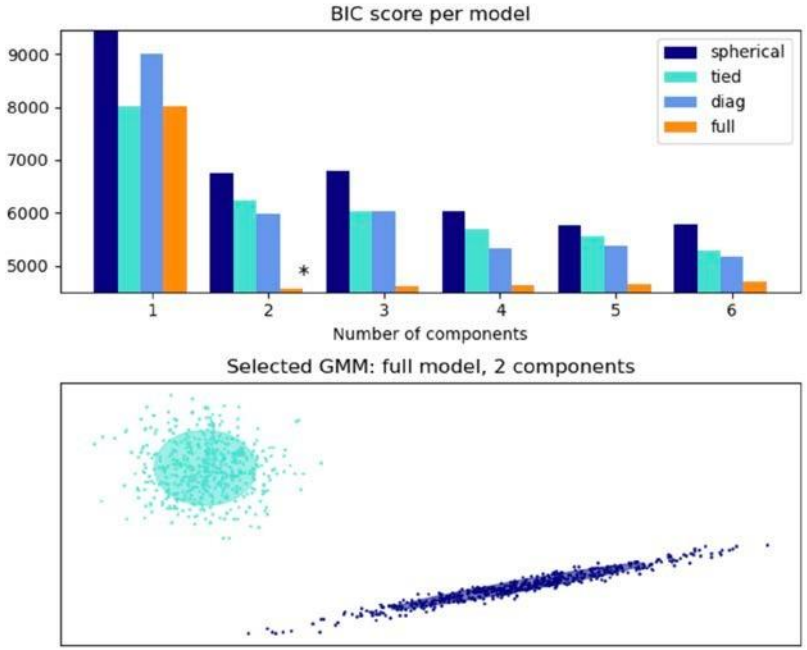
- As we increase K, the likelihood will keep increasing
- More mixtures -> more parameters -> overfits



http://staffblogs.le.ac.uk/bayeswithstata/2014/05/22/mixture-models-how-many-components/

Picking the amount of Gaussians

- Need a measure of goodness (like Elbow method in k- mean)
- Bayesian Information Criterion (BIC)
- Penalize the log likelihood from the data by the number of parameters in the model
  - -2 log L + t log (n)
  - t = number of parameters in the model
  - n = number of data points
- We want to minimize BIC



https://scikit-learn.org/stable/auto_examples/mixture/plot_gmm_selection.html

Latent variables?

- Discrete latent variables: clusters/partitions data into subgroups
- Continuous latent variables: can be used for dimensionality reduction

Image segmentation with GMM EM

- D - {r,g,b} value at each pixel
- Latent : segment where each pixel comes from
- Hyperparameters: number of mixtures (K), initial values

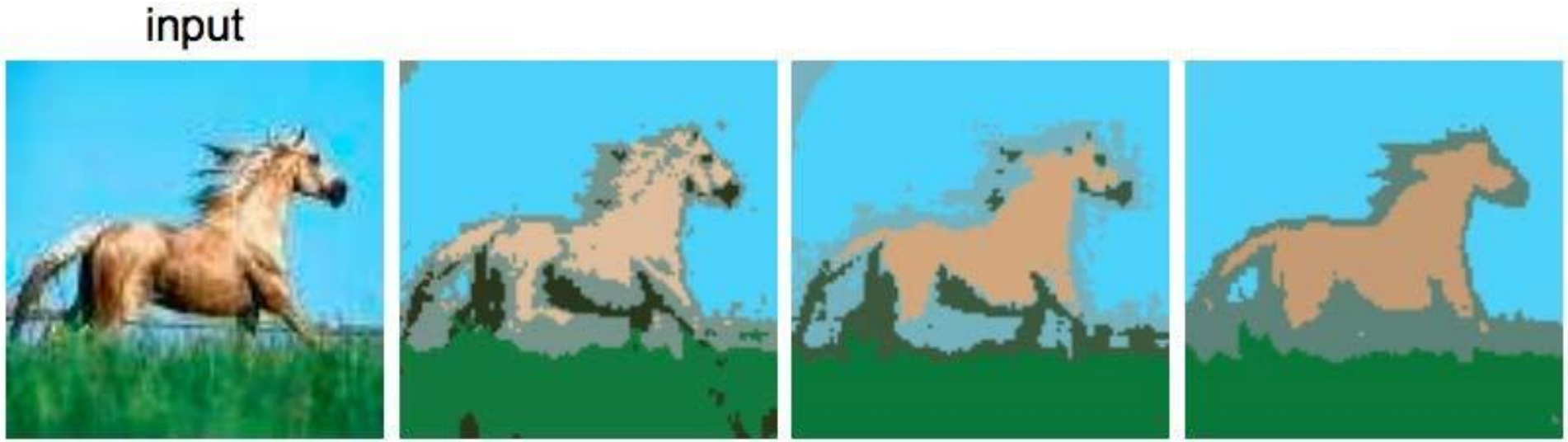


Image segmentation with GMM EM

(a) (b) (c)

Fig. 1. Original images: (a) flower, (b) tiger, (c) bear

Fig. 2. Segmentation results ($M = 2$)

Fig. 3. Segmentation results ($M = 5$)

Zhaoxia Fu Color Image Segmentation Using Gaussian Mixture Model and EM Algorithm

Language/Topic modeling

THE UNITED STATES CONSTITUTION

Latent variable:
Topic
P(word|topic)

For examples: see Probabilistic latent semantic analysis

Summary

- GMM
  - Mixture of Gaussians
- EM
  - Expectation
  - Maximization

More info and exact proofs
http://cs229.stanford.edu/summer2019/cs229-notes8.pdf